CloudOps CI/CD Monitor: End-to-End DevOps Pipeline on AWS

End-to-End DevOps Pipeline on AWS

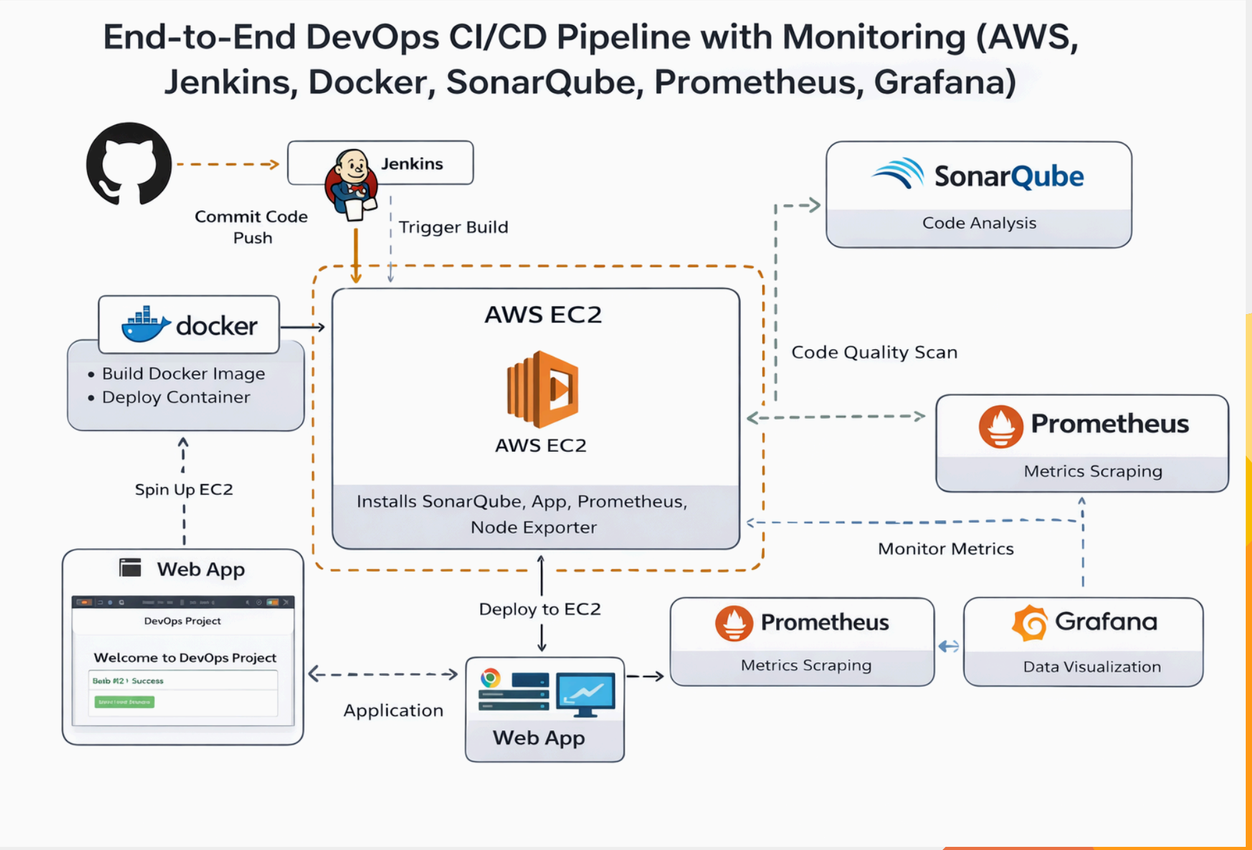
By: Vaishnavi Gandhre

Project Overview

- This project demonstrates a complete DevOps CI/CD pipeline that automates code integration, quality analysis, containerized deployment, and real-time monitoring.

Project Workflow

- 1. Developer pushes code to GitHub
- 2. Jenkins triggers CI/CD pipeline
- 3. SonarQube performs code quality analysis
- 4. Docker builds the application image
- 5. Application deployed on AWS EC2
- 6. Prometheus collects system metrics
- 7. Grafana visualizes monitoring dashboards



Tools and Roles

- GitHub – Version control & source code
- Jenkins – CI/CD automation
- SonarQube – Static code quality analysis
- Docker – Containerization for consistent environments
- AWS EC2 – Cloud infrastructure hosting
- Prometheus – Metrics collection
- Grafana – Monitoring dashboards

Problems Solved

- Eliminates manual deployments
- Improves code quality and security
- Ensures consistent environments
- Enables faster releases
- Provides real-time system monitoring

Key Features

- Automated CI/CD pipeline
- Code quality gate checks
- Container-based deployment
- Cloud-hosted infrastructure
- Monitoring and visualization dashboards

Conclusion

This DevOps pipeline demonstrates how automation, monitoring, and cloud infrastructure can work together to deliver reliable and scalable applications.

Thank You